TO SOLVE DIFFERENTIAL EQUATION USING FROBENIAS METHOD.

Find a regular singular point of the following differential equation.

(a) $2x^2 \frac{d^2y}{dx^2} + 3x\frac{dy}{dx} + (x^2-4)y = 0$.

(b) $x^2(x-2)^2y'' + 2(x-2)y' + (x+3)y = 0$,

SOLUTION.

(a). $\frac{d^2y}{dx^2} + \frac{3x}{2x^2}\frac{dy}{dx} + \frac{(x^2-4)}{2x^2}y = 0$,

$P(x) = \frac{3x}{2x^2} = \frac{3}{2x}$ $\quad Q(x) = \frac{x^2-4}{2x^2}$

$\Phi(x) = 0$.

at $x = 0$.

$P(0) = \infty \quad Q(0) = \infty \quad \Phi(0) = 0$.

The point $x=0$ is a regular singular point

Using Frobenias method.

Case 1. when roots λ_1 and λ_2 are district and not differing by integers.

$\lambda_1 - \lambda_2 = 0$ or

$\lambda_1 = \lambda_2 \quad \lambda_2 = 2$.

$\lambda_1 - \lambda_2 = \lambda_2 - 2 = -3/2$

The complete solution is given by

$y(x) = c_1(y)_{\lambda_1} + c_2(y)_{\lambda_2}$

case 2: when λ_1 and λ_2 are equals

$\lambda_1 = \lambda_2$

The complete solution is given by

$$y(x) = c_1(y)_{\lambda_1} + c_2\left(\frac{\partial y}{\partial \lambda}\right)_{\lambda_2}$$

case 3: when roots λ_1 and λ_2 are distinct and differ by integer ($\lambda_1 < \lambda_2$)

$\lambda_1 = 3/2 \quad \lambda_2 = 5/2$.

$\lambda_1 - \lambda_2 = 3/2 - 5/2 = -1$

Example find the solution of the following differential equation.

$$3x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 0 \quad \text{--- (i)}$$

Solution.

$$\frac{d^2y}{dx^2} + \frac{2}{3x}\frac{dy}{dx} + \frac{1}{3x}y = 0$$

$P(x) = 2/3x \quad Q(x) = 1/3x \quad \phi(x) = 0$.

at $x = 0$,

$P(0) = \infty \quad Q(0) = \infty \quad \phi(0) = 0$.

The point is a regular singular point.

Step 2: let. $y(x) = \sum_{k=0}^{\infty} a_k x^{\lambda+k}$ ——— (2)

$$y'(x) = \sum_{k=0}^{\infty} (\lambda+k) a_k x^{\lambda+k-1} \quad \text{——— (3)}$$

$$y''(x) = \sum_{k=0}^{\infty} (\lambda+k)(\lambda+k-1) a_k x^{\lambda+k-2} \quad \text{——— (4)}$$

Step 3: use (2), (3), and (4) in (1) to get:

$$3x \frac{d^2y}{dx^2} + 2\, dy/dx + y = 0.$$

$$3x \sum_{k=0}^{\infty} (\lambda+k)(\lambda+k-1) a_k x^{\lambda+k-2} + 2\sum_{k=0}^{\infty} (\lambda+k) a_k x^{\lambda+k-1} + \sum_{k=0}^{\infty} a_k x^{\lambda+k} = 0.$$

$$\sum_{k=0}^{\infty} 3(\lambda+k)(\lambda+k-1) a_k x^{\lambda+k-1} + \sum_{k=0}^{\infty} 2(\lambda+k) a_k x^{\lambda+k-1} + \sum_{k=0}^{\infty} a_k x^{\lambda+k} = 0.$$

Step 4: Equating the coefficient of x on L.H.S and R.H.S.
The indicial equation we put $k=0$,

$$\sum_{k=0}^{\infty} 3\lambda(\lambda-1) a_0 x^{\lambda-1} + \sum_{k=0}^{\infty} 2\lambda a_0 x^{\lambda-1} + \sum_{k=0}^{\infty} a_0 x^{\lambda} = 0.$$

$$3a_0(\lambda-1)\lambda + 2\lambda a_0 = 0$$

$$(3(\lambda-1)\lambda + 2\lambda) a_0 = 0$$

$3\lambda^2 - 3\lambda + 2\lambda = 0$.

$\boxed{3\lambda^2 - \lambda = 0}$ — indicial equation.

$\lambda = 0$ or $3\lambda - 1 = 0$.

$\lambda_1 = 0$ or $\lambda_2 = 1/3$.

Step 5: To generate the general formula we put $k = 1$.

$$\sum_{k=0}^{\infty} 3(\lambda+k)(\lambda+k-1) a_k x^{\lambda+k-1} + \sum_{k=0}^{\infty} 2(\lambda+k) a_k x^{\lambda+k-1} + \sum_{k=0}^{\infty} a_k x^{\lambda+k} = 0.$$

$k = k + 1$

$$\sum_{k=0}^{\infty} 3(\lambda+k)(\lambda+k-1) a_k$$

$$\sum_{k=0}^{\infty} 3(\lambda+k+1)(\lambda+k) a_{k+1} x^{\lambda+k} + \sum_{k=0}^{\infty} 2(\lambda+k+1) a_{k+1} x^{\lambda+k} + \sum_{k=0}^{\infty} a_k x^{\lambda+k} = 0.$$

$$= 3a_{k+1}(\lambda+k+1)(\lambda+k) + 2a_{k+1}(\lambda+k+1) + a_k = 0.$$

$$\left[3(\lambda+k+1)(\lambda+k) + 2(\lambda+k+1)\right] a_{k+1} + a_k = 0.$$

$$\boxed{a_{k+1} = \frac{-a_k}{(\lambda+k+1)\left(3(\lambda+k)+2\right)}}$$

$k = 0$, $a_1 = \dfrac{-a_0}{(\lambda+1)(3\lambda+2)}$.

$k=1$. $a_2 = \dfrac{-a_1}{(\lambda+2)(3\lambda+5)}$.

$$\therefore a_2 = \frac{a_0}{(\lambda+1)(3\lambda+2)(\lambda+2)(3\lambda+5)}.$$

$k=2$.

$$a_3 = \frac{-a_2}{(\lambda+3)(3\lambda+8)}.$$

Now for $\lambda_1=0$, we have

$$a_1 = -a_0/2.$$

$$a_2 = \frac{1}{20}a_0.$$

$$a_3 = -\frac{1}{480}a_0.$$

$$y_1(x) = [a_0+a_1x+a_2x^2+\dots]x^{\lambda_1}$$

$$y_1(x) = a_0\left(1-\tfrac{1}{2}x+\tfrac{1}{20}x^2+\dots\right)x^0$$

When $\lambda_2 = 1/3$ we get

$$a_1 = -\tfrac{1}{4}a_0.$$

$$a_2 = \tfrac{1}{56}a_0. \quad a_3 = -\frac{1}{1?0}a_0.$$

$$y_2(x) = \left[a_0 + a_1 x + a_2 x^2 + \cdots\right] x^{\lambda_2}$$

$$y_2(x) = a_0 \left(1 - \tfrac{1}{4} x + \tfrac{1}{56} x^2 + \cdots\right) x^{1/3}.$$

$$y(x) = c_1 y_1(x)_{\lambda_1} + c_2 y_2(x)_{\lambda_2}$$

$$y(x) = a_0\left(1 - \tfrac{1}{2} x + \tfrac{1}{20} x^2 + \cdots\right) + a_0\left(1 - \tfrac{1}{4} x + \tfrac{1}{56} x^2 + \cdots\right) x^{1/3}$$